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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

- Lists Part 1 : Introduction (unit?unit=57&lesson=58)
- Lists Part 2 : Manipulation (unit?unit=57&lesson=59)
- Lists Part 3 : Operations (unit?unit=57&lesson=60)
- Lists Part 4 : Slicing (unit?unit=57&lesson=61)
- Loops and Conditionals : Fizzbuzz 01 (unit?unit=57&lesson=62)
- Loops and Conditionals : Fizzbuzz 02 (unit?unit=57&lesson=63)
- Crowd Computing - Just estimate 01 (unit?unit=57&lesson=64)
- Crowd Computing - Just estimate 02 (unit?unit=57&lesson=65)
- Crowd Computing - Just estimate 03 (unit?unit=57&lesson=66)
- Crowd Computing - Just estimate 04 (unit?unit=57&lesson=67)
- Crowd Computing - Just estimate 05 (unit?unit=57&lesson=68)
- Crowd Computing - Just estimate 06 (unit?

Week 3 : Assignment 3

The due date for submitting this assignment has passed.

Due on 2025-02-12, 23:59 IST.

Assignment submitted on 2025-02-12, 21:28 IST

1) Which of the following is/are **false** statement(s)? ***1 point***

- [x] Lists are fixed in size, they cannot store more than 1024 values
- [x] One can access element in list by using non-numeric indices
- [ ] Iterating over lists is possible in Python
- [x] We need to specify required size of list while creating a new list variable

Yes, the answer is correct.
Score: 1

Accepted Answers:
Lists are fixed in size, they cannot store more than 1024 values
One can access element in list by using non-numeric indices
We need to specify required size of list while creating a new list variable

2) In the below code -

```
n= ?
a = []
for i in range(n):
    l1 = []
    p = 0
    for j in range(i):
        l1.append(j)
    for k in l1:
        p+=1
    a.append(p)
k=0
for l in a:
    k+=l
print(k)
```

For what value of **n** does the program print **10** ?

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 5

1 point

3) From the previous question, for what values of **n** is the number **25** appended to list a. ***1 point***

- [ ] 25
- [ ] 24
- [x] 35
- [x] 32

unit=57&lesson=69)
- Permutations - Jumbled Words 01 (unit?unit=57&lesson=70)
- Permutations - Jumbled Words 02 (unit?unit=57&lesson=71)
- Permutations - Jumbled Words 03 (unit?unit=57&lesson=72)
- Theory of Evolution 01 (unit?unit=57&lesson=73)
- Theory of Evolution 02 (unit?unit=57&lesson=74)
- Theory of Evolution 03 (unit?unit=57&lesson=75)
- Theory of Evolution 04 (unit?unit=57&lesson=76)
- Week 3 Feedback Form : The Joy of Computing using Python!! (unit?unit=57&lesson=77)
- **Quiz: Week 3 : Assignment 3 (assessment?name=508)**
- Week 3: Programming Assignment 1 (/noc25_cs69/progassignment?name=510)
- Week 3: Programming Assignment 2 (/noc25_cs69/progassignment?name=511)
- Week 3: Programming Assignment 3 (/noc25_cs69/progassignment?name=513)

week 4 ()

Week 5 ()

Week 6 ()

Week 7 ()

Week 8 ()

Week 9 ()

Week 10 ()

Week 11 ()

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Problem Solving Session - Jan 2025 ()

Yes, the answer is correct.
Score: 1
Accepted Answers:
35
32

4) What does the following code perform ? ***1 point***

```
def mystery(container):
  result = []
  for i in range(len(container)):
    if i % 2 != 0:
      result.append(container[i] * 4)
    else:
      result.append(container[i] * (-1))
  return result
```

- It converts any input list into a new list which is filled with some numbers which are always in increasing order
- It converts any input list into a new list such that at even indices, the value is a multiple of an even number and at odd indices, the value is either even or negative number
- It converts any input list into a new list such that at odd indices, the value is always even number and at even indices, the value is negative
- It converts any input list into a new list, which follows no pattern

Yes, the answer is correct.
Score: 1
Accepted Answers:
It converts any input list into a new list such that at odd indices, the value is always even number and at even indices, the value is negative

5) From the previous question, if the option - ***1 point***

It converts any input list into a new list such that at even indices, the value is a multiple of an even number and at odd indices, the value is either even or negative number.

is incorrect, Can you make changes to code such that it this option is always true for any input ?

- No, it is not possible to make such changes
- No, the option is already correct
- Yes, we can make changes

Yes, the answer is correct.
Score: 1
Accepted Answers:
Yes, we can make changes

6) If **file.txt** exists, Does the code successfully run ? ***1 point***

```
with open("file.txt", "r") as f:
  data = f.read()
  print(data)
```

- Yes
- No

Yes, the answer is correct.
Score: 1
Accepted Answers:
Yes

7) Which of the following are examples of Social Computing ? ***1 point***

- ChatGPT
- Wikipedia
- Quora
- Twitch

No, the answer is incorrect.
Score: 0
Accepted Answers:
Wikipedia
Quora

8) What does the following code print for **n = 256**? ***1 point***

```
n = ?
k = 0
a = []
b = 0
while (n != 0):
  k = k + (n % 10)
  a.append(n % 10)
  n = n // 10
a.sort()
for i in a:
  b = b + i
if(b != k):
  print("PYTHON IS FUN")
else:
  print("PYTHON IS BORING")
```

- PYTHON IS BORING
- PYTHON IS FUN

Yes, the answer is correct.
Score: 1

Accepted Answers:
PYTHON IS BORING

9) From the previous question, is **a.sort()** responsible for printing of either of the two possible sentences ? ***1 point***

- Yes, and the code may print ”PYTHON IS FUN”
- No, it is purely dependent on variable n
- No, it is not dependent on a.sort(), and the code will never print ”PYTHON IS FUN”

Yes, the answer is correct.
Score: 1

Accepted Answers:
No, it is not dependent on a.sort(), and the code will never print ”PYTHON IS FUN”

10) What does the code in question 8, calculate ?

Can you say what the values of k and b are if n = 245326785456 x 10^{3359} .

If values of k and b are different, enter 0, else enter value of k.

0

No, the answer is incorrect.
Score: 0

Accepted Answers:
(Type: Numeric) 57

1 point